

Project: MerchantPay

Payment Gateway & Merchant Acquiring Platform

Domain: Banking & Financial Services

1. Introduction

MerchantPay is a web-based payment gateway and merchant acquiring platform designed for acquiring banks, payment service providers, and fintech companies to manage merchant onboarding, terminal and virtual POS provisioning, transaction authorisation workflows, settlement processing, chargeback and dispute management, fraud monitoring, and merchant analytics. The system enables merchants to track their transaction activity and settlement status; acquiring operations teams to manage merchant lifecycle and terminal deployments; dispute teams to handle chargeback cases and retrieval requests; fraud teams to monitor suspicious transaction patterns; and administrators to configure merchant categories, fee schedules, and risk rules.

It uses a REST API-based backend architecture with Angular/React frontend and is compatible with Java (Spring Boot) and .NET (ASP.NET Core) platforms.

Actors / Users

- **Merchant:** Views transaction history, settlement statements, dispute status, and device inventory
- **Merchant Onboarding Officer:** Manages merchant applications, KYB verification, and terminal provisioning
- **Acquiring Operations Analyst:** Monitors transaction pipelines, settlement runs, and exception queues
- **Dispute Analyst:** Manages chargeback cases, retrieval requests, and representment workflows
- **Fraud Analyst:** Reviews flagged transactions, investigates suspicious patterns, and manages rule thresholds
- **Acquiring Admin:** Configures MCC categories, interchange fee schedules, settlement cycles, and risk rules

2. Module Overview

- **2.1 Identity & Access Management:** Authentication, RBAC, and audit trails.
- **2.2 Merchant Onboarding & KYB Management:** Manages merchant applications, business verification, and account activation workflows.
- **2.3 Terminal & Device Management:** Manages POS terminal provisioning, deployment tracking, and device lifecycle.
- **2.4 Transaction Processing & Authorisation:** Records and tracks payment transaction lifecycles from authorisation to capture.
- **2.5 Settlement & Funding Management:** Manages daily settlement runs, merchant funding disbursements, and reconciliation.
- **2.6 Chargeback & Dispute Management:** Handles chargeback initiations, retrieval requests, and representment case workflows.
- **2.7 Fraud Monitoring & Risk Management:** Flags suspicious transactions, manages fraud rules, and tracks investigations.
- **2.8 Notifications & Alerts:** Transaction alerts, settlement credit confirmations, chargeback deadlines, and fraud flag notifications.

3. Architecture Overview

- **Frontend:** Angular or React
- **Backend:** REST API-based architecture
- **Database:** Relational Database (MySQL / PostgreSQL / SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features

- Merchant and acquirer staff registration with role-based and merchant-scoped access control
- Full audit logging for all transaction overrides, chargeback decisions, and fee adjustments

Entities

- **User**

- UserID
- Name
- Role (Merchant/OnboardingOfficer/AcqOps/DisputeAnalyst/FraudAnalyst/Admin)
- Email
- Phone
- MerchantID
- Status (Active/Locked/Inactive)
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - EntityType
 - RecordID
 - Timestamp

4.2 Merchant Onboarding & KYB Management Module

Features

- Process merchant applications with business registration, ownership, and bank account details
- Manage KYB document verification, risk categorisation, and account activation workflows

Entities

- **MerchantApplication**
 - AppID
 - BusinessName
 - RegistrationNumber
 - MCC
 - BusinessType
 - OwnerName

- NationalIDRef
- SettlementBankAccountRef
- ApplicationDate
- Status (Submitted/KYBInProgress/Approved/Rejected/Active)
- **Merchant**
 - MerchantID
 - AppID
 - MerchantCode
 - BusinessName
 - MCC
 - RiskCategory (Low/Medium/High)
 - SettlementFrequency (Daily/Weekly/Monthly)
 - FeeScheduleID
 - Status (Active/Suspended/Terminated)
- **KYBDocument**
 - DocID
 - MerchantID
 - DocumentType
(BusinessRegistration/DirectorID/BankLetter/TaxCertificate/AddressProof)
 - FilePath
 - UploadedDate
 - VerificationStatus (Pending/Verified/Rejected)

4.3 Terminal & Device Management Module

Features

- Provision physical and virtual POS terminals to merchants with serial and configuration details
- Track terminal deployment status, maintenance history, and decommissioning

Entities

- **Terminal**
 - TerminalID
 - MerchantID
 - TerminalType (PhysicalPOS/mPOS/VirtualPOS/PaymentLink/QR)
 - SerialNumber
 - Model
 - ProvisionedDate
 - LocationDescription
 - Status (Active/Inactive/UnderMaintenance/Decommissioned)
- **TerminalEvent**
 - EventID
 - TerminalID
 - EventType (Deployment/Swap/Repair/SoftwareUpdate/Decommission)
 - EventDate
 - PerformedByID
 - Notes
 - Status (Completed/Pending)

4.4 Transaction Processing & Authorisation Module

Features

- Record payment transactions across card, wallet, and QR payment channels
- Track full transaction lifecycle from authorisation through capture, void, and refund

Entities

- **PaymentTransaction**
 - TxnID
 - MerchantID
 - TerminalID
 - PaymentChannel (Card/Wallet/QR/NetBanking/UPI)

- TransactionType (Purchase/Refund/Void/PreAuth/Capture)
- Amount
- Currency
- CardBIN
- AuthorisationCode
- TransactionDateTime
- Status (Authorised/Captured/Declined/Voided/Refunded/Pending)
- **TransactionBatch**
 - BatchID
 - MerchantID
 - TerminalID
 - BatchDate
 - TotalTransactions
 - TotalAmount
 - Status (Open/Closed/Submitted/Settled)

4.5 Settlement & Funding Management Module

Features

- Run daily settlement processing for closed batches and compute merchant net funding amounts
- Track interchange, scheme fees, acquirer markup, and merchant fee deductions

Entities

- **SettlementRun**
 - SettlementID
 - SettlementDate
 - ProcessedByID
 - TotalMerchantsSettled
 - TotalGrossAmount
 - TotalFees

- TotalNetFunded
- Status (Draft/Processing/Completed/Failed)
- **MerchantSettlement**
 - MerchantSettlementID
 - SettlementID
 - MerchantID
 - GrossAmount
 - InterchangeFee
 - SchemeFee
 - AcquirerMarkup
 - NetFunding
 - FundingBankRef
 - Status (Pending/Funded/Failed)

4.6 Chargeback & Dispute Management Module

Features

- Log chargeback cases and retrieval requests received from card schemes
- Manage representment workflows, evidence compilation, and resolution tracking

Entities

- **ChargebackCase**
 - CaseID
 - TxnID
 - MerchantID
 - ReasonCode
 - ClaimAmount
 - ReceivedDate
 - ResponseDeadline
 - AssignedAnalystID

- Status (Received/DocumentsRequested/EvidenceSubmitted/Won/Lost/Reversed)
- **DisputeEvidence**
 - EvidenceID
 - CaseID
 - EvidenceType
(TransactionReceipt/DeliveryProof/CustomerCorrespondence/SignedSlip/AuthLog)
 - FilePath
 - UploadedDate
 - Status (Submitted/Accepted/Insufficient)

4.7 Fraud Monitoring & Risk Management Module

Features

- Flag transactions matching configurable fraud detection rules
- Manage investigation workflows, merchant risk scoring, and rule threshold updates

Entities

- **FraudFlag**
 - FlagID
 - TxnID
 - MerchantID
 - FraudType
(VelocityBreach/CardTestingPattern/HighDeclineRate/AnomalousAmount/Suspicious MCC)
 - RiskScore
 - FlaggedDate
 - AssignedAnalystID
 - Status (Open/UnderInvestigation/Cleared/Confirmed/Reported)
- **MerchantRiskScore**
 - ScoreID
 - MerchantID
 - ComputedDate

- RiskScore
- KeyFactors
- Category (Low/Medium/High/Critical)
- ReviewRecommendation
- Status (Current/Superseded)

4.8 Notifications & Alerts Module

Features

- Transaction decline alerts, settlement credit confirmations, and chargeback receipt notifications for merchants
- Fraud flag escalations, high decline rate warnings, and chargeback response deadline alerts for operations teams

Entities

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Transaction/Settlement/Chargeback/Fraud/Terminal/Compliance)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB
- **Production:** Cloud or on-premises with secure APIs and stateless services

6. Database Design

Tables:

- **User**(UserID, Name, Role, Email, Phone, MerchantID, Status)

- **AuditLog**(AuditID, UserID, Action, EntityType, RecordID, Timestamp)
- **MerchantApplication**(AppID, BusinessName, RegistrationNumber, MCC, BusinessType, OwnerName, NationalIDRef, SettlementBankAccountRef, ApplicationDate, Status)
- **Merchant**(MerchantID, AppID, MerchantCode, BusinessName, MCC, RiskCategory, SettlementFrequency, FeeScheduleID, Status)
- **KYBDocument**(DocID, MerchantID, DocumentType, FilePath, UploadedDate, VerificationStatus)
- **Terminal**(TerminalID, MerchantID, TerminalType, SerialNumber, Model, ProvisionedDate, LocationDescription, Status)
- **TerminalEvent**(EventID, TerminalID, EventType, EventDate, PerformedByID, Notes, Status)
- **PaymentTransaction**(TxnID, MerchantID, TerminalID, PaymentChannel, TransactionType, Amount, Currency, CardBIN, AuthorisationCode, TransactionDateTime, Status)
- **TransactionBatch**(BatchID, MerchantID, TerminalID, BatchDate, TotalTransactions, TotalAmount, Status)
- **SettlementRun**(SettlementID, SettlementDate, ProcessedByID, TotalMerchantsSettled, TotalGrossAmount, TotalFees, TotalNetFunded, Status)
- **MerchantSettlement**(MerchantSettlementID, SettlementID, MerchantID, GrossAmount, InterchangeFee, SchemeFee, AcquirerMarkup, NetFunding, FundingBankRef, Status)
- **ChargebackCase**(CaseID, TxnID, MerchantID, ReasonCode, ClaimAmount, ReceivedDate, ResponseDeadline, AssignedAnalystID, Status)
- **DisputeEvidence**(EvidenceID, CaseID, EvidenceType, FilePath, UploadedDate, Status)
- **FraudFlag**(FlagID, TxnID, MerchantID, FraudType, RiskScore, FlaggedDate, AssignedAnalystID, Status)
- **MerchantRiskScore**(ScoreID, MerchantID, ComputedDate, RiskScore, KeyFactors, Category, ReviewRecommendation, Status)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Merchant Portal:** Transaction dashboard, settlement statement, chargeback case tracker, terminal inventory, fee breakdown
- **Onboarding Console:** Merchant application queue, KYB document review, risk categorisation, activation workflow

- **Operations Dashboard:** Transaction volume heatmap, batch status board, settlement run monitor, exception queue
- **Dispute Workbench:** Chargeback case list, evidence upload form, deadline tracker, representment submission
- **Fraud Monitoring Console:** Flagged transaction queue, risk score dashboard, rule threshold configurator, investigation log
- **Admin Console:** MCC catalog, fee schedule builder, settlement cycle configurator, fraud rule manager

8. Non-Functional Requirements

- **Performance:** Handle 10,000+ concurrent merchant and operations users with high-volume transaction data ingestion
- **Security:** Card data masking (BIN only stored), RBAC, chargeback evidence access control, and full audit trails
- **Scalability:** Multi-acquirer and multi-currency support for payment service providers across geographies
- **Availability:** 99.95% uptime for transaction processing and settlement modules critical to merchant cash flow
- **Maintainability:** Configurable fraud rules, fee schedules, chargeback reason code mappings, and settlement cycle rules
- **Observability:** Real-time transaction pipeline dashboards, decline rate monitors, and chargeback rate alerts

9. Assumptions & Constraints

- Phase 1 excludes integration with card scheme networks (Visa/Mastercard), payment switches, and banking payment APIs
- Transaction data is entered via batch file upload or manual entry; live payment gateway API integration is deferred to Phase 2
- All notifications are in-app only; merchant email and SMS alerts are out of Phase 1 scope
- Fraud rules are configured internally; machine learning-based scoring models are deferred
- Fully implementable using standard Java Spring Boot / ASP.NET Core, Angular/React, and RDBMS tooling

